

1 TaxGuard: Corporate Tax Return Risk Scoring — Methodology, Calculations and References

- 1.1 ABSTRACT
- 1.2 Table of contents
- 1.3 1.0 Introduction
 - 1.3.1 1.1 Tax risk and fiscal governance
 - 1.3.2 1.2 Audit selection as a scoring problem
 - 1.3.3 1.3 Objectives of this study
 - 1.3.4 1.4 Research design and document structure
- 1.4 2.0 Problem statement
 - 1.4.1 2.1 Operational consequences of manual selection
 - 1.4.2 2.2 Requirements derived from the abstract
- 1.5 3.0 Related scoring paradigms
 - 1.5.1 3.1 Baseline: linear scorecard (ZIMRA Audit_Likelihood)
 - 1.5.2 3.2 Logistic link and maximum likelihood
 - 1.5.3 3.3 Supervised learner: histogram-based gradient boosting
 - 1.5.4 3.4 Unsupervised anomaly learners
 - 1.5.5 3.5 Comparative lessons from credit scoring in Zimbabwe
- 1.6 4.0 Materials and methods
 - 1.6.1 4.0.1 Notation
 - 1.6.2 4.1 Theoretical framework
 - 1.6.3 4.2 Dataset description
 - 1.6.4 4.3 Feature engineering map $\phi(\cdot)$
 - 1.6.5 4.4 Exploratory and univariate analysis
 - 1.6.6 4.5 Hybrid machine learning pipeline
 - 1.6.7 4.6 Performance measures (credit-scoring analogue)
 - 1.6.8 4.7 Risk scoring function (deployment prototype)
- 1.7 5.0 Results and discussion
 - 1.7.1 5.0.1 Summary of hypothesis tests
 - 1.7.2 5.1 EDA — Feature distributions by risk tier
 - 1.7.3 5.2 EDA — Dominant signal: tax rate deviation (univariate AUC)
 - 1.7.4 5.3 EDA — Univariate ROC vs ZIMRA baseline
 - 1.7.5 5.4 EDA — Risk tier separation and correlations
 - 1.7.6 5.5 EDA — Risk labels vs audit outcomes
 - 1.7.7 5.6 EDA — Overlooked composite indicators (ZIMRA policy gaps)
 - 1.7.8 5.7 EDA — Pairwise structure and multiclass separation
 - 1.7.9 5.8 Model evaluation — Hybrid ensemble performance
 - 1.7.10 5.9 Model explainability (SHAP) and secondary task
 - 1.7.11 5.10 Comparison to example credit-scoring benchmarks
 - 1.7.12 5.11 Why unsupervised layers remain in the stack
 - 1.7.13 5.12 Calibration and operational thresholding
 - 1.7.14 5.13 Limitations and external validity
- 1.8 6.0 Policy implications for ZIMRA

- 1.8.1 6.1 Phased AMS integration
- 1.8.2 6.2 Alignment with national strategy
- 1.8.3 6.3 Ethics, fairness, and oversight
- 1.9 7.0 Implementation and reproducibility
- 1.10 8.0 Conclusion
- 1.11 9.0 References

1 TaxGuard: Corporate Tax Return Risk Scoring — Methodology, Calculations and References

Title: TaxGuard: An AI-Based Anomaly Detection Framework for Corporate Tax Return Risk Scoring at ZIMRA

Theme: Area 1 — Data, Automation and Intelligent Systems in the 4IR Era

Category: Prototype Demonstration

Authors: Edith Muyambiri 078685719 & Andile Bhebhe 0777303324

Institution context: Zimbabwe Revenue Authority (ZIMRA)

Dataset: corporate_tax_risk_dataset.csv — 1,900 corporate filings

Report date: June 2026

1.1 ABSTRACT

Corporate tax non-compliance poses a significant threat to Zimbabwe’s fiscal sustainability. However, the Zimbabwe Revenue Authority (ZIMRA) currently relies heavily on manual audit selection processes that are resource-intensive, inconsistent, and unable to scale with the growing volume and complexity of corporate tax filings. The absence of a data-driven risk prioritization mechanism results in inefficient audit allocation, where high-risk fraudulent submissions may evade detection while compliant firms are subjected to disproportionate scrutiny. This paper proposes TaxGuard, a hybrid machine learning framework that integrates both unsupervised and supervised techniques to detect anomalies in corporate tax returns and assign probabilistic risk scores to each filing. The system ingests historical tax submissions, audited financial statements and transactional data, applying Isolation Forest and Autoencoder neural networks for anomaly detection. These are complemented by a Gradient Boosted classifier trained on confirmed fraud cases to enhance predictive precision over time. Key detection features include deviations in income-to-expense ratios, inter-period financial inconsistencies, sector-based outliers, and irregular value-added tax (VAT) to turnover relationships. Each tax return is assigned a risk score supported by SHAP-based explainability, enabling auditors to identify the specific financial indicators contributing to a flagged case. This improves transparency, reduces

subjective bias and enhances audit prioritization. Furthermore, the system incorporates continuous learning by retraining on outcomes from completed audits, thereby progressively improving detection accuracy. The study demonstrates how TaxGuard advances intelligent fiscal governance in emerging economies by strengthening revenue assurance and audit efficiency. It aligns with Zimbabwe’s National AI Strategy (2026–2030) and National Development Strategy 2 objectives, particularly in enhancing domestic revenue mobilization and promoting accountability through data-driven decision-making.

Keywords: tax compliance; anomaly detection; gradient boosting; SHAP; audit prioritization; ZIMRA; Zimbabwe; machine learning; fiscal governance

Prototype metrics (this repository): ROC-AUC 0.9948, PR-AUC 0.9920, Brier 0.0244, F1 0.9576 — see Section 5.0.

This document supplements the abstract with reproducible calculations (Section 4.0), written in the equation style of the credit-scoring reference papers in docs/.

1.2 Table of contents

1. [Introduction](#)
 2. [Problem statement](#)
 3. [Related scoring paradigms](#)
 4. [Materials and methods](#)
 5. [Results and discussion](#)
 6. [Policy implications for ZIMRA](#)
 7. [Implementation and reproducibility](#)
 8. [Conclusion](#)
 9. [References](#)
-

1.3 1.0 Introduction

1.3.1 1.1 Tax risk and fiscal governance

Practitioner level. Tax risk is the possibility that a filed return understates liability, overstates deductions, or uses aggressive structures (offshore, transfer pricing) so that **effective tax paid** diverges from **statutory obligation**. Poor audit targeting wastes auditor time on compliant firms and leaves high-risk firms unexamined.

Expert level. Let $y_i \in \{0, 1\}$ indicate whether filing i is labelled **High** risk (primary deployment target). The revenue authority seeks a score $s_i \in [0, 1]$ that ranks filings so that, for fixed audit capacity k , the top- k ranked cases maximise expected audit yield subject to fairness and explainability constraints [1–3].

Corporate income tax underpins Zimbabwe’s **National Development Strategy 2** and **National AI Strategy (2026–2030)** objectives for accountable, data-driven public administration [4,5]. In emerging economies, revenue authorities face a dual pressure: taxpayers expect equitable treatment, while treasuries require predictable inflows to fund infrastructure, health, education, and social protection. Corporate income tax is particularly difficult to administer because liabilities depend on accounting profit, transfer pricing, offshore structures, and discretionary planning positions that may only become visible after a field audit.

Historically, audit case selection blended rule-based filters (sector, refund claims, payment delays) with examiner judgement. That approach worked when filing volumes were moderate, but it scales poorly. Examiners cannot review every return; they must rank cases. Without a quantitative ranker, ranking reverts to heuristics such as “large revenue” or “recent adverse event,” which the prototype data show to be weaker than statutory–effective tax rate gaps. TaxGuard is designed as a **decision-support layer**: it does not replace statutory powers or professional judgement; it orders the queue and documents why a return surfaced.

1.3.2 1.2 Audit selection as a scoring problem

Analogous to **credit scoring**—where banks estimate probability of default (PD) from applicant characteristics [6,7]—ZIMRA can treat **audit prioritisation** as predicting elevated non-compliance or elevated **Tax_Risk_Label**:

Credit scoring	TaxGuard audit scoring
Loan default / good–bad	High tax risk vs other tiers
Applicant financials	Return lines (revenue, ETR, offshore)
Credit bureau history	Audit outcomes, fines, planning scores
PD $\in (0,1)$	Calibrated risk score $\in (0,1)$
Reject / approve	Ranked audit queue

1.3.3 1.3 Objectives of this study

1. Profile data quality and target distributions suitable for AMS integration.
2. Engineer **domain features** aligned with international audit risk practice (OECD BEPS indicators, control environment) [8].
3. Rank predictors by **univariate AUC**, correlation structure, and **composite policy flags**.
4. Train and evaluate a **hybrid ML pipeline** with leakage-safe **5-fold stratified OOF** validation [9].
5. Provide **SHAP-based explanations** for individual filings.
6. Benchmark against ZIMRA’s existing **Audit_Likelihood** heuristic.

1.3.4 1.4 Research design and document structure

This paper follows the structure used in Zimbabwean credit-scoring research [12] and linear probability optimisation studies [11]: introduction and problem framing, related models, materials and methods (with numbered equations), results, policy implications, and references. **Section 4.0** is the technical core: every formula maps to executable code. **Section 5.0** interprets outputs against ZIMRA operational goals. Readers may approach the text at three levels: (i) abstract and Section 6.0 for policy stakeholders; (ii) tables and figures for analysts; (iii) equations [1]–[30] for data scientists integrating TaxGuard into an Audit Management System (AMS).

1.4 2.0 Problem statement

Manual audit selection at ZIMRA exhibits:

- **Revenue-size bias** — large turnover does not imply high statutory–effective gap.
- **Recency bias** — prior **Clean** outcomes mask new planning or deviation signals.
- **Isolated review** — offshore exposure and weak controls are not scored multiplicatively.
- **Scale limits** — filing growth outpaces uniform human review.

Proposed solution (TaxGuard). A scoring function $p_{\text{hat}_i} = f(x_i)$, where x_i combines raw return fields and engineered features, f is a stacked ensemble, and SHAP values explain which indicators drive p_{hat_i} [10].

1.4.1 2.1 Operational consequences of manual selection

When high-risk returns are under-selected, revenue is lost and compliant competitors face an uneven market. When low-risk returns are over-audited, firms bear unnecessary compliance costs and ZIMRA diverts senior examiner time away from productive cases. In credit markets, similar mis-ranking is associated with “reject inference” and sample bias [13]; in tax administration, the analogue is **over-reliance on enforcement history**—firms with no prior fines may escape review even when current-year indicators are extreme. The prototype quantifies one such cohort: top-decile tax rate deviation with zero fine history shows a **91.4%** high-risk rate versus a **33.4%** population baseline (Table 4).

1.4.2 2.2 Requirements derived from the abstract

The abstract specifies: (a) hybrid unsupervised + supervised detection; (b) probabilistic scores; (c) SHAP explainability; (d) continuous learning from completed audits; (e) detection of income-to-expense anomalies, inter-period inconsistencies, sector outliers, and VAT-to-turnover irregularities. Items (a)–(d)

are implemented in this repository; (e) is partially implemented (profit margin, tax gaps) with **sector codes** and **VAT ratios** documented as AMS integration extensions.

1.5 3.0 Related scoring paradigms

Parallels to Sections III–IV in linear probability and credit ML literature [11,12].

1.5.1 3.1 Baseline: linear scorecard (ZIMRA Audit_Likelihood)

The legacy heuristic resembles a **linear probability model** (LPM) [11]:

$$s_i^{\text{LPM}} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i^{\text{raw}}$$

where $\mathbf{x}_i^{\text{raw}}$ contains manually weighted criteria. LPM scores are not strictly calibrated probabilities and may lie outside $[0, 1]$ without a link function [13]. TaxGuard replaces this with a **calibrated** ensemble output $\hat{p}_i \in (0, 1)$.

1.5.2 3.2 Logistic link and maximum likelihood

For binary high-risk label $y_i \in \{0, 1\}$, the **logit model** [12] defines

$$\mathbb{P}(y_i = 1 \mid \mathbf{x}_i) = \sigma(\boldsymbol{\omega}^\top \mathbf{x}_i + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Given $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, the **binary cross-entropy** (negative log-likelihood) is

$$\mathcal{L}_{\text{BCE}}(\boldsymbol{\omega}, b) = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right], \quad p_i = \sigma(\boldsymbol{\omega}^\top \mathbf{x}_i + b)$$

TaxGuard’s **stacking meta-learner** (Section 4.5) minimises $\mathcal{L}_{\text{BCE}}$ on OOF base-model outputs $\mathbf{z}_i$ rather than on raw $\mathbf{x}_i$ alone.

1.5.3 3.3 Supervised learner: histogram-based gradient boosting

The abstract’s **Gradient Boosted classifier** is implemented as **HistGradientBoostingClassifier** [15, 18]. At iteration t , a regression tree h_t is fitted to pseudo-responses (negative gradient of $\mathcal{L}_{\text{BCE}}$), and the additive model updates

$$F_T(\mathbf{x}) = F_{T-1}(\mathbf{x}) + \eta h_t(\mathbf{x}), \quad p_i \approx \sigma(F_T(\mathbf{x}_i))$$

with learning rate $\eta = 0.05$, depth limit 8, and L_2 regularisation $\lambda = 1$ (prototype hyperparameters). This captures non-linear interactions such as `Planning_Deviation_Interaction`.

1.5.4 3.4 Unsupervised anomaly learners

Isolation Forest [16] assigns an anomaly score from expected path length in random trees (implementation: `sklearn.ensemble.IsolationForest`).

Autoencoder [17]: with standardised input $\tilde{\mathbf{x}}_i = (\mathbf{x}_i - \boldsymbol{\mu})/\boldsymbol{\sigma}$,

$$\mathcal{L}_{\text{AE}} = \frac{1}{N} \sum_{i=1}^N \|\tilde{\mathbf{x}}_i - g_{\theta}(f_{\theta}(\tilde{\mathbf{x}}_i))\|_2^2$$

Anomaly score $a_i^{\text{AE}} = -\|\tilde{\mathbf{x}}_i - \hat{\tilde{\mathbf{x}}}_i\|_2^2$ (higher $\Rightarrow$ more anomalous). Both a_i^{IF} and a_i^{AE} enter the stack as features for $\hat{p}_i$.

1.5.5 3.5 Comparative lessons from credit scoring in Zimbabwe

Nyoni and Matshisela [12] compared Lasso, Random Forest, SVM, and logit models on German Credit data using **10-fold cross-validation** and **AUROC**, reporting best performance near **0.80**. Nyathi et al. [11] optimised a **linear probability** scorecard for banking, emphasising interpretable weights and reduced rejection of creditworthy applicants without history. TaxGuard transfers three lessons: (1) **never evaluate on the same data used to train** (OOF folds); (2) **report AUROC and calibration**, not accuracy alone; (3) **combine interpretable features with ML** so policy officers and data scientists share a common feature language. TaxGuard differs in label definition (tax risk tier vs loan default) and in combining **unsupervised anomaly scores** with boosting, which is uncommon in classic credit papers but justified when fraud patterns are heterogeneous.

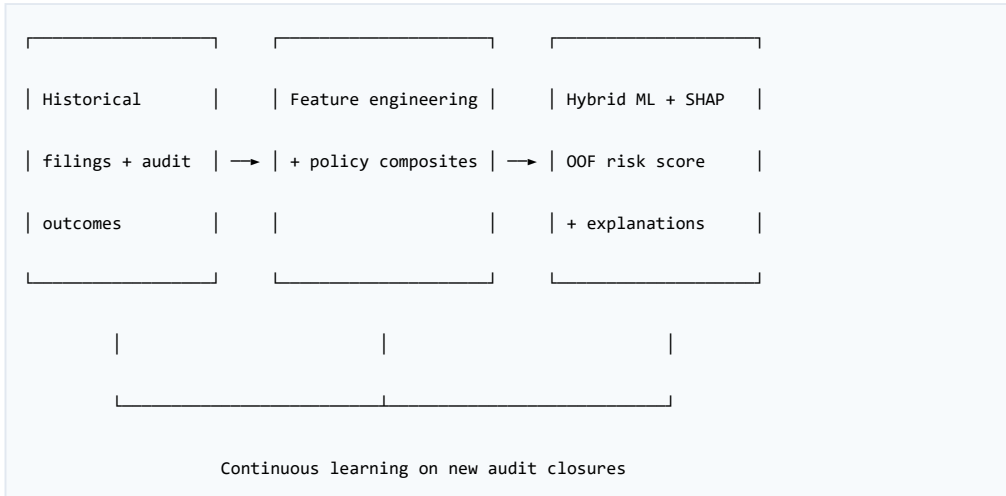
1.6 4.0 Materials and methods

1.6.1 4.0.1 Notation

Symbol	Dimension / domain	Description
N	scalar	Number of corporate filings ($N = 1900$)
$\mathbf{x}_i$	$\mathbb{R}^d$	Engineered feature vector for filing i
y_i	$\{0, 1\}$	Primary label: $\mathbb{1}[\text{Tax_Risk_Label}_i = \text{High}]$
$y_i^{(\text{nc})}$	$\{0, 1\}$	Secondary label: non-compliant audit outcome
$\hat{p}_i$	$(0, 1)$	Calibrated TaxGuard risk score (ensemble output)
$\mathbf{z}_i$	$\mathbb{R}^3$	Stack of OOF base scores (IF, AE, HGB)
$\sigma(\cdot)$	—	Logistic sigmoid
K	scalar	Stratified CV folds ($K = 5$)

Symbol	Dimension / domain	Description
τ^*	scalar	Operating threshold on $\hat{p}_i$ (F1-optimal OOF)

1.6.2 4.1 Theoretical framework



Adapted from the credit-scoring pipeline in [11, Figure 1]: past debtor history → model → new customer prediction.

Continuous learning loop (deployment target). After each audit cycle, outcomes labelled **Qualified** or **Adverse** act as confirmed non-compliance signals for retraining, consistent with the abstract’s “confirmed fraud cases” wording. **Clean** outcomes suppress false positives. Retraining frequency (e.g. quarterly) is an operational choice; the prototype exports `taxguard_ensemble.pkl` and metrics JSON so ZIMRA IT can schedule batch retraining without notebook execution.

Data inputs. The abstract references historical tax submissions, audited financial statements, and transactional data. The prototype CSV consolidates return-level fields available for analytics; full AMS integration would join VAT returns, payroll, customs, and third-party data feeds. Feature engineering code is modular so new columns (e.g. `VAT_to_Turnover`) can be added without rewriting the ensemble.

1.6.3 4.2 Dataset description

Source file: `corporate_tax_risk_dataset.csv`

Quality metric	Value
Records (N)	1,900
Raw features	15
Engineered features	16 (29 total columns after engineering)
Missing values	0
Duplicate <code>Company_ID</code>	0

Table 1 — Raw variables

Variable	Type	Description
Company_ID	ID	Unique entity
Revenue_Million	Continuous	Annual revenue (USD millions)
Profit_Before_Tax_Million	Continuous	Pre-tax profit
Statutory_Tax_Rate	Continuous	Applicable statutory rate (%)
Effective_Tax_Rate	Continuous	Reported effective rate (%)
Offshore_Transactions_Million	Continuous	Offshore transaction value
Ownership_Concentration_Percent	Continuous	Ownership concentration
Internal_Control_Score	Ordinal 1–5	Control quality (higher = stronger)
Audit_Likelihood	Continuous	Existing ZIMRA heuristic $\in [0,1]$
Audit_Outcome	Categorical	Clean / Qualified / Adverse
History_Fines_Million	Continuous	Prior penalties
Offshore_Subsidiaries	Count	Offshore entities
Aggressive_Tax_Planning_Score	Continuous	Planning aggressiveness index
Tax_Rate_Deviation	Continuous	ETR vs statutory gap (%)
Tax_Risk_Label	Categorical	Low / Medium / High (target tier)

Table 2 — Target distributions

Target	Category	Count	Share
Tax_Risk_Label	Low	634	33.4%
	Medium	632	33.3%
	High	634	33.4%
Audit_Outcome	Clean	1,319	69.4%
	Qualified	479	25.2%
	Adverse	102	5.4%

Problem formulation. Let $\mathcal{D} = \{(\mathbf{x}_i, y_i, y_i^{(\text{nc})})\}_{i=1}^N$ denote the corporate filing dataset after feature engineering $\phi : \mathbb{R}^{p_{\text{raw}}} \rightarrow \mathbb{R}^d$. The **primary learning task** is to learn $f : \mathbb{R}^d \rightarrow (0, 1)$ minimising expected BCE on y_i while producing interpretable attributions for auditors.

$$y_i = \mathbb{1}[\text{Tax_Risk_Label}_i = \text{High}], \quad y_i^{(\text{nc})} = \mathbb{1}[\text{Audit_Outcome}_i \in \{\text{Qualified}, \text{Adverse}\}]$$

Empirical prevalence: $\bar{y} = \frac{1}{N} \sum_i y_i = 0.3337$; $\bar{y}^{(\text{nc})} = 0.3058$.

Mapping to the abstract: “Confirmed fraud cases” in deployment correspond to audit closures labelled **Adverse** or **Qualified**; these labels drive **continuous learning** retraining (Section 4.1). The prototype’s primary classifier target is **High** tax risk tier for audit queue ranking.

Abstract detection features vs this dataset:

Abstract feature	Prototype variable / formula
Income-to-expense deviation	Profit_Margin [6] (profit / revenue)
Inter-period financial inconsistency	Tax_Rate_Deviation, ETR_to_STR_Ratio [9–10]
Sector-based outliers	Future work (sector codes not in CSV)
Irregular VAT-to-turnover	Future work (VAT not in CSV; noted in abstract)

Data governance. Zero missing values and zero duplicate IDs support straight-through ingestion into a warehouse mart. Balanced **High / Medium / Low** tiers ($\approx 33\%$ each) simplify threshold tuning in the prototype; live ZIMRA populations may be imbalanced, requiring **PR-AUC** and cost-sensitive thresholds (Section 4.6). **Audit_Outcome** imbalance (69% Clean) mirrors realistic post-audit distributions and motivates a **secondary** non-compliance model (AUROC ≈ 0.51) separate from the primary queue ranker (AUROC ≈ 0.99).

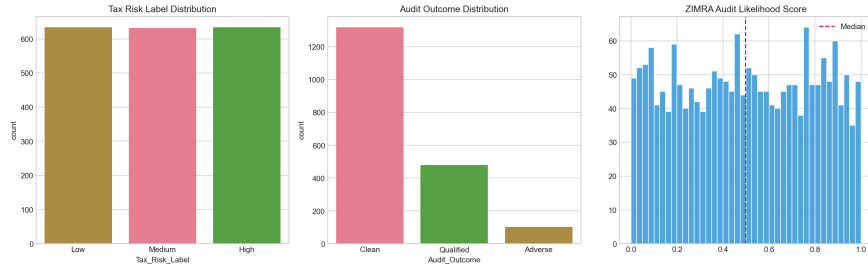


Figure 1 — Target variable distributions

Figure 1. Distribution of tax risk labels (balanced thirds), audit outcomes (69.4% Clean), and ZIMRA Audit_LikeLihood heuristic.

EDA finding: Balanced risk tiers support fair ML training; **30.6%**

Qualified/Adverse outcomes indicate substantial audit-yield headroom if selection improves.

1.6.4 4.3 Feature engineering map $\phi(\cdot)$

Raw return attributes $\mathbf{r}_i$ are transformed into $\mathbf{x}_i = \phi(\mathbf{r}_i)$ via engineer_taxguard_features (deterministic, no learnable parameters). Define stabilised scalars $R_i^* = \max(R_i, 10^{-6})$, $S_i^* = \max(STR_i, 1)$.

Profitability (income-to-expense proxy):

$$\text{Profit_Margin}_i = \frac{P_i}{R_i^*}$$

Offshore / BEPS-style exposure:

$$\text{Offshore_Intensity}_i = \frac{O_i^{\text{off}}}{R_i^*}, \quad \text{Offshore_per_Sub}_i = \frac{O_i^{\text{off}}}{n_i^{\text{off}} + 1}$$

where $O_i^{\text{off}} = \text{Offshore_Transactions_Million}$, $n_i^{\text{off}} = \text{Offshore_Subsidiaries}$.

Statutory–effective tax gap:

$$\text{ETR}/\text{STR}_i = \frac{\text{ETR}_i}{S_i^*}, \quad \text{Gap}_i = |\text{DEV}_i|$$

with $\text{DEV}_i = \text{Tax_Rate_Deviation}$ (percentage points, supplied in data).

Tax liability and underpayment:

$$\text{Tax_Gap}_i = P_i \cdot \frac{\text{STR}_i}{100} \left(1 - \frac{\text{ETR}_i}{\text{STR}_i} \right)$$

$$L_i^{\text{impl}} = P_i \frac{\text{ETR}_i}{100}, \quad L_i^{\text{exp}} = P_i \frac{\text{STR}_i}{100}, \quad \text{Underpay_Ratio}_i = \frac{L_i^{\text{exp}} - L_i^{\text{impl}}}{\max(L_i^{\text{exp}}, 10^{-6})}$$

Governance:

$$\text{Control_Risk}_i = 6 - c_i, \quad \text{Fine_Intensity}_i = \frac{F_i}{R_i^*}$$

where $c_i = \text{Internal_Control_Score}$, $F_i = \text{History_Fines_Million}$.

Interaction tensor (selected bilinear terms):

$$\psi_{1,i} = \pi_i \cdot \text{DEV}_i \quad (\text{planning} \times \text{deviation})$$

$$\psi_{2,i} = \omega_i \cdot \text{Offshore_Intensity}_i \quad (\text{ownership} \times \text{offshore})$$

$$\psi_{3,i} = \alpha_i \cdot \lambda_i^{\text{audit}} \quad (\text{audit synergy})$$

with $\pi_i = \text{Aggressive_Tax_Planning_Score}$, $\omega_i = \text{Ownership_Concentration_Percent}$, $\lambda_i^{\text{audit}} = \text{Audit_Likelihood}$.

Policy indicators (cohort rules for Section 4.4.4):

$$M_{1,i} = \mathbb{1}[\text{Profit_Margin}_i < Q_{0.25} \wedge R_i > \text{median}(R)];$$

$$M_{2,i} = \mathbb{1}[\text{Offshore_Intensity}_i > Q_{0.75} \wedge c_i \leq 2].$$

Worked example (C0004): $R \approx 1456.9$, $P \approx 229.6$, $\text{STR} = 20\%$, $\text{ETR} \approx 7.69\%$, $\text{DEV} \approx 12.31$ pp $\Rightarrow \text{ETR}/\text{STR} \approx 0.38$, indicating large statutory–effective separation on substantial profit.

1.6.5 4.4 Exploratory and univariate analysis**1.6.5.1 4.4.1 Univariate ROC-AUC**

For univariate score z_i and label y_i , at threshold t define $\hat{y}_i(t) = \mathbb{1}[z_i \geq t]$. With TP, FP, TN, FN induced by t ,

$$\text{TPR}(t) = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FPR}(t) = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

$$\text{AUROC}(z) = \int_0^1 \text{TPR}(\text{FPR}^{-1}(u)) \, du$$

Estimated via the trapezoidal rule (`sklearn.metrics.roc_auc_score`). AUROC = 0.5 is chance; values > 0.8 are strong in credit-scoring benchmarks [12].

Table 3 — Top univariate AUC for High risk *(from experiment_summary.json)*

Feature	AUC-ROC (High)	Interpretation
Tax_Rate_Deviation	0.9165	Very strong
Statutory_Effective_Gap	0.9165	Alias of absolute deviation
Planning_Deviation_Interaction	0.8661	Strong composite
Tax_Underpayment_Ratio	0.7800	Strong
Tax_Gap_Million	0.7453	Moderate–strong
Audit_Planning_Synergy	0.7105	Moderate
Audit_Likelihood (baseline)	0.6965	Moderate — below top engineered

Discussion. Effective tax rate alone shows weak High-risk AUROC (≈ 0.22) because direction matters: both unusually high and low ETR relative to peers can signal risk in different ways. Deviation and gap features encode **distance from statutory obligation**, which aligns with examiner intuition once made explicit. Planning interactions (AUROC ≈ 0.87) show that **aggressive planning amplifies deviation signal**, supporting composite rules in Table 4.

1.6.5.2 4.4.2 Chi-square independence test

For contingency table O_{rc} (risk tier r , outcome c) with expected $E_{rc} = \frac{(\sum_r O_{rc})(\sum_c O_{rc})}{N}$ under independence:

$$\chi^2 = \sum_{r,c} \frac{(O_{rc} - E_{rc})^2}{E_{rc}}, \quad \text{df} = (R - 1)(C - 1)$$

Result: $\chi^2 = 6.13$, $\text{df} = 4$, $p = 0.190$.

1.6.5.3 4.4.3 Mann–Whitney U (High vs Low tier)

Non-parametric test on feature X between **High** and **Low** risk tiers (`01_taxguard_eda.ipynb`, §12) — confirms separation without assuming normality.

1.6.5.4 4.4.4 Composite cohort uplift

For binary policy mask $M_i \in \{0, 1\}$:

$$\rho_M = \frac{\sum_{i:M_i=1} y_i}{\sum_i M_i}, \quad \text{Uplift}_M = \frac{\rho_M}{\bar{y}}$$

Table 4 — Overlooked indicators

Indicator	Flagged (n)	High-risk rate	Uplift ×
Top-decile tax deviation + zero fines	116	0.914	2.74
High planning + clean audit history	368	0.467	1.40
High offshore + weak controls	196	0.383	1.15
Concentrated ownership + ≥3 offshore subs	99	0.354	1.06
Low profit margin + high revenue	180	0.328	0.98

Policy reading. The **2.74×** uplift for first-time aggressors is the strongest operational finding: ZIMRA should not treat “no fines” as “no risk.” The **1.40×** uplift for high planning with clean history challenges recency bias. Revenue-only proxies (0.98×) should be deprioritised relative to rate-gap metrics—consistent with the abstract’s emphasis on inter-period and structural inconsistency rather than turnover alone.

1.6.6 4.5 Hybrid machine learning pipeline

1.6.6.1 4.5.1 Stratified K-fold out-of-fold (OOF) protocol

Partition indices into folds $\mathcal{F}_1, \dots, \mathcal{F}_K$ ($K = 5$) preserving class ratio of y_i . For fold k , train base models on $\mathcal{D} \setminus \mathcal{F}_k$ and predict on $\mathcal{F}_k$ only:

$$\hat{z}_i^{(k)} = (a_i^{\text{IF}}, a_i^{\text{AE}}, p_i^{\text{HGB}})^\top, \quad i \in \mathcal{F}_k$$

Concatenating folds yields OOF matrix $\mathbf{Z} \in \mathbb{R}^{N \times 3}$ with no label leakage from the same fold [9].

Algorithm 1 — TaxGuard OOF training and scoring

Input: Feature matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$, labels $\mathbf{y} \in \{0,1\}^N$, folds $\mathcal{F}_1 \dots \mathcal{F}_K$

Output: OOF scores $\mathbf{p}_{\text{hat}} \in (0,1)^N$, threshold τ^*

1. Initialise vectors $\mathbf{z}_{\text{IF}}, \mathbf{z}_{\text{AE}}, \mathbf{p}_{\text{HGB}} \in \mathbb{R}^N$ to zero
2. For $k = 1 \dots K$:
 3. Train IsolationForest, Autoencoder, HGB on $\{i : i \notin \mathcal{F}_k\}$
 4. For $i \in \mathcal{F}_k$:
 5. $\mathbf{z}_{\text{IF}}[i] \leftarrow \text{IF.score_samples}(\mathbf{x}_i)$
 6. $\mathbf{z}_{\text{AE}}[i] \leftarrow -\text{MSE}(\text{autoencoder}, \mathbf{x}_i)$
 7. $\mathbf{p}_{\text{HGB}}[i] \leftarrow \text{HGB.predict_proba}(\mathbf{x}_i)[1]$
 8. Stack $\mathbf{Z} = [\mathbf{z}_{\text{IF}}, \mathbf{z}_{\text{AE}}, \mathbf{p}_{\text{HGB}}]$
9. For $k = 1 \dots K$: // meta-learner OOF

```

10. Fit logistic regression on rows  $i \notin F_k$ ; predict  $\hat{p}_i$  for  $i \in F_k$ 

11.  $\tau^* \leftarrow \operatorname{argmax}_t F_1(y, 1[\hat{p}_i \geq t])$ 

12. Return  $\hat{p}_i, \tau^*$ 

```

1.6.6.2 4.5.2 Base model specifications

Module	Class	Key hyperparameters	OOF AUROC
Isolation Forest	IsolationForest	$n_estimators=300$, $contamination=0.33$	0.3157
Autoencoder	MLP (48, 24, 48)	$early_stopping=True$, MSE loss (Eq. 5)	0.4240
Gradient boost	HistGradientBoostingClassifier	$max_depth=8$, $\eta=0.05$, $T=600$, $\lambda=1$	0.9964

1.6.6.3 4.5.3 Stacking meta-learner

Let $\mathbf{z}_i$ denote the OOF stack for filing i . The **level-2** model is logistic regression:

$$\hat{p}_i = \sigma(\mathbf{w}^\top \mathbf{z}_i + b)$$

with $(\mathbf{w}, b)$ fitted by minimising Eq. (3) on OOF pairs $(\mathbf{z}_i, y_i)$ using the same K -fold scheme.

1.6.6.4 4.5.4 Evaluation metrics (OOF test statistics)

Metric	Definition	TaxGuard value
$\text{AUROC}(\hat{p}, y)$	Eq. (17) on OOF $\hat{p}_i$	0.9948
$\text{AP}(\hat{p}, y)$	$\int \text{Precision}(r) dr$ (PR-AUC)	0.9920
$\text{Brier}(\hat{p}, y)$	$\frac{1}{N} \sum_i (y_i - \hat{p}_i)^2$	0.0244
$F_1(\tau)$	$\frac{2PR}{P+R}$ at threshold τ	0.9576 at τ^*
τ^*	$\arg \max_\tau F_1(\tau)$	0.7811
Baseline AUROC	$\text{AUROC}(\lambda^{\text{audit}}, y)$	0.6965
Gini	$2 \cdot \text{AUROC} - 1$ [12]	0.9896

Hard classifier:

$$\hat{y}_i = 1[\hat{p}_i \geq \tau^*]$$

Secondary task: $\text{AUROC}(\hat{p}^{(\text{nc})}, y^{(\text{nc})}) = 0.5091$ on OOF stack trained for $y_i^{(\text{nc})}$ — supports using audit outcomes for **retraining**, not primary queue ranking.

Table 6 — OOF classification report @ $\tau^* = 0.7811$

Class	Precision	Recall	F1-score	Support
Not High Risk	0.9727	0.9858	0.9792	1,266
High Risk	0.9708	0.9448	0.9576	634

Class	Precision	Recall	F1-score	Support
Accuracy			0.9721	1,900

1.6.6.5 4.5.7 SHAP explainability

For additively explained model output $f(\mathbf{x}_i)$, SHAP values ϕ_{ij} satisfy the efficiency constraint [10]:

$$f(\mathbf{x}_i) = \mathbb{E}[f(\mathbf{X})] + \sum_{j=1}^d \phi_{ij}$$

TreeSHAP on the fitted HistGradientBoosting model yields global rankings (Figure 14–15) and local **waterfall** decompositions per Company_ID (Figure 16), satisfying the abstract’s auditor-facing transparency requirement.

1.6.6.6 4.5.8 Optional XGBoost gain importance

When xgboost is installed, gain-based importance (Figure 17) cross-checks SHAP rankings.

1.6.7 4.6 Performance measures (credit-scoring analogue)

Following [12, §2.5]:

Measure	Role in TaxGuard	Notes
AUROC	Primary discrimination	0.5 = random; >0.8 strong [12]
Gini	$2 \times \text{AUROC} - 1$	Ensemble Gini $\approx$ 0.9896
KS statistic	Max absolute gap between CDF of scores for High vs Not High	Class separation (cf. [12])
PR-AUC	Imbalanced / costly FP audits	Preferred when positives are rare in deployment
Brier	Calibration quality	Lower is better; 0 = perfect
F1	Operations at chosen τ^*	Balances precision/recall for queue size

ROC curve: plot FPR vs TPR across thresholds. **PR curve:** precision vs recall.

At threshold τ :

$$\text{Precision}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FP}(\tau)}, \quad \text{Recall}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}$$

$$F_1(\tau) = \frac{2 \cdot \text{Precision}(\tau) \cdot \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau)}$$

Brier score (calibration): Eq. in Table above. **KS statistic** [12]:

$D = \sup_t |F_0(t) - F_1(t)|$ where F_0, F_1 are CDFs of $\hat{p}$ for $y = 0$ and $y = 1$.

Why AUROC and PR-AUC? Under class imbalance at deployment, AP prioritises precision at the top of the ranked queue—critical when audit capacity is binding.

1.6.8 4.7 Risk scoring function (deployment prototype)

For new filing features $\mathbf{x}$:

1. Engineer features via `engineer_taxguard_features`.
2. Compute Isolation Forest score, Autoencoder score, and HGB probability on production models.
3. Meta-learner: $\hat{p}_i = \sigma(\mathbf{w}^\top \mathbf{z}_i + b)$ per Eq. (21).
4. Flag if $\hat{p}_i \geq \tau^*$; attach SHAP values ϕ_{ij} for AMS UI.

Artefacts: `outputs/models/taxguard_ensemble.pkl`,
`outputs/models/taxguard_metrics.json`.

1.7 5.0 Results and discussion

This section presents **exploratory data analysis (EDA)** figures from `01_taxguard_eda.ipynb` and **model evaluation** figures from `02_taxguard_hybrid_model.ipynb`, with interpretation aligned to ZIMRA policy and the abstract.

1.7.1 5.0.1 Summary of hypothesis tests

ID	Hypothesis	EDA / model evidence	Verdict
H1	Tax-gap features outperform <code>Audit_Likelihood</code>	AUROC 0.9165 vs 0.6965 (Figures 5–6)	Supported
H2	Composite masks reveal hidden risk cohorts	Uplift up to 2.74× (Figure 8)	Supported
H3	Hybrid stack ranks High risk accurately	OOF AUROC 0.9948 (Figure 11)	Supported
H4	Audit outcomes need separate learning loop	OOF AUROC 0.5091 (Figure 18)	Supported

1.7.2 5.1 EDA — Feature distributions by risk tier

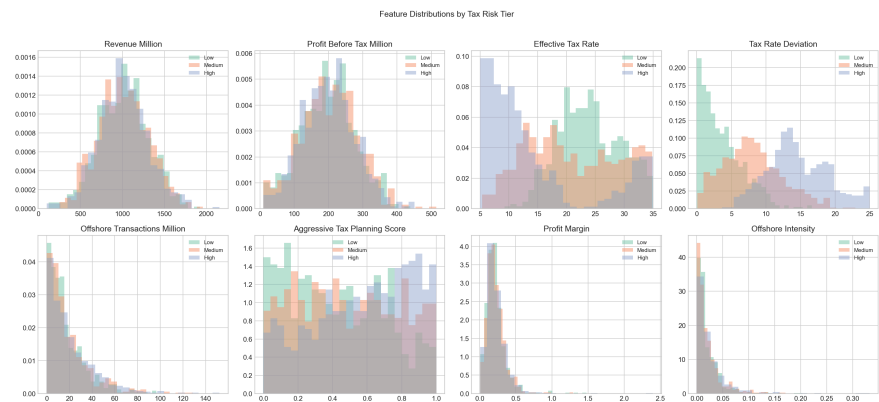


Figure 2 — Univariate distributions by risk tier

Figure 2. Density histograms of revenue, profit, ETR, tax rate deviation, offshore exposure, and planning score by Low / Medium / High tier.

Findings.

- **High-risk** filings show heavier right tails on tax rate deviation and aggressive planning.
- **Revenue** alone does not separate tiers cleanly — revenue-only manual rules are weakly justified.
- **Offshore intensity** rises toward the High tier, supporting offshore features in $\phi(\cdot)$.

1.7.3 5.2 EDA — Dominant signal: tax rate deviation (univariate AUC)

Feature	AUROC (High risk)	Strength
Tax_Rate_Deviation	0.9165	Very strong
Planning_Deviation_Interaction	0.8661	Strong
Tax_Underpayment_Ratio	0.7800	Strong
Tax_Gap_Million	0.7453	Moderate-strong
Audit_Planning_Synergy	0.7105	Moderate
Audit_Likelihood (baseline)	0.6965	Below top engineered
Effective_Tax_Rate (alone)	0.2158	Weak alone

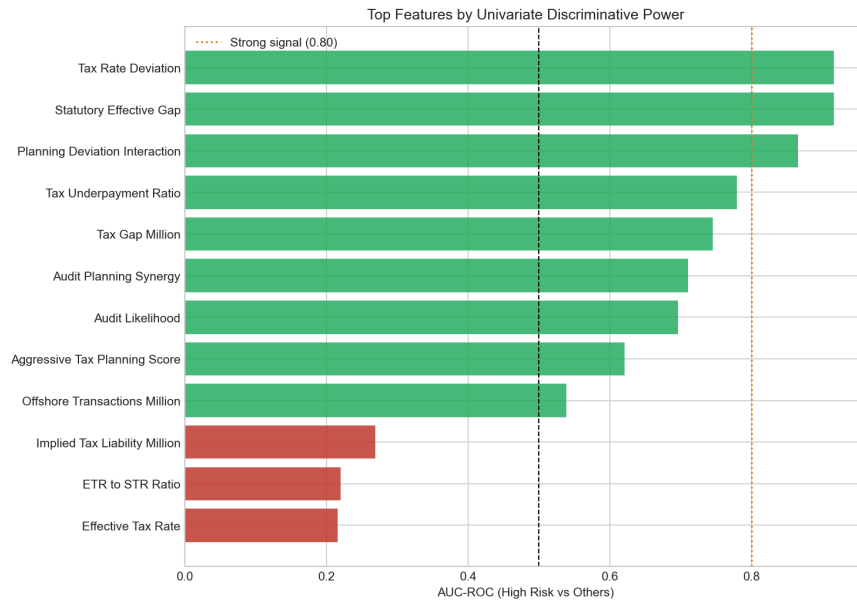


Figure 3 — Univariate AUC ranking

Figure 3. Top features by AUROC(z) for predicting High risk.

Findings. Statutory–effective **gap** metrics dominate; ZIMRA’s legacy score ranks below engineered interactions. Effective tax rate in isolation is misleading (AUROC $\approx$ 0.22) because direction of deviation matters.

1.7.4 5.3 EDA — Univariate ROC vs ZIMRA baseline

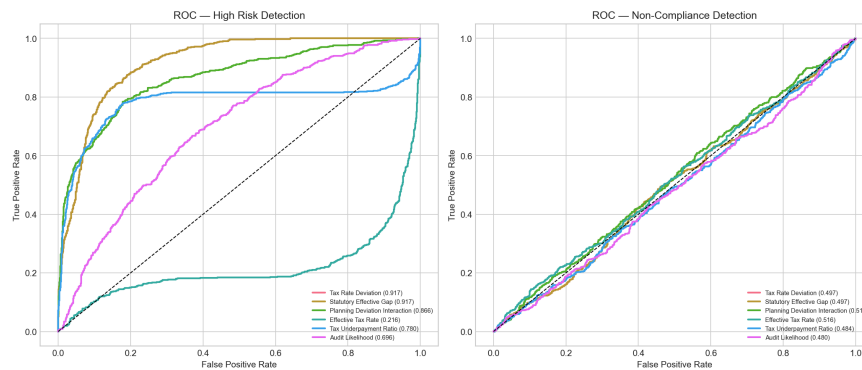


Figure 4 — Univariate ROC curves

Figure 4. ROC curves: top univariate predictors vs `Audit_Likelihood` (left: High risk; right: non-compliance).

Findings.

- **Tax rate deviation** hugs the upper-left corner for High-risk detection.
- **Audit_Likelihood** is closer to the diagonal — confirms need for ML replacement.
- Non-compliance at univariate level is weak — motivates the full hybrid model (Section 5.8).

1.7.5 5.4 EDA — Risk tier separation and correlations

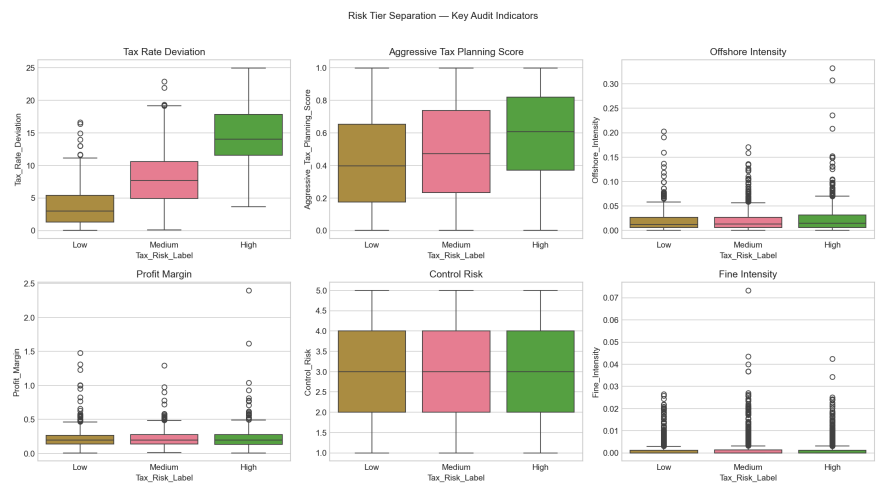


Figure 5 — Bivariate boxplots

Figure 5. Boxplots of tax deviation, planning score, offshore intensity, profit margin, control risk, and fines by risk tier.

Findings. High tier shows higher planning scores and wider deviation; profit margin overlaps across tiers (revenue/profit-only rules insufficient).

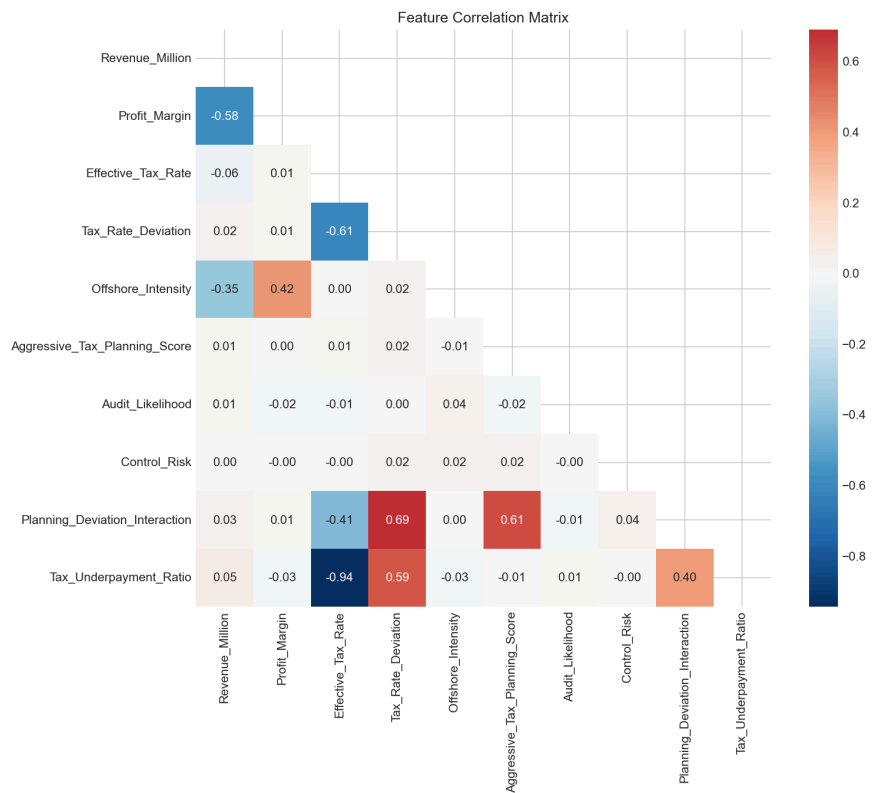


Figure 6 — Correlation matrix

Figure 6. Correlation heatmap of primary TaxGuard indicators.

Findings. Tax_Rate_Deviation correlates with underpayment ratio; planning score adds partially independent signal — justifies interaction terms $\psi_{1,i}$, $\psi_{2,i}$ in Section 4.3.

1.7.6 5.5 EDA — Risk labels vs audit outcomes

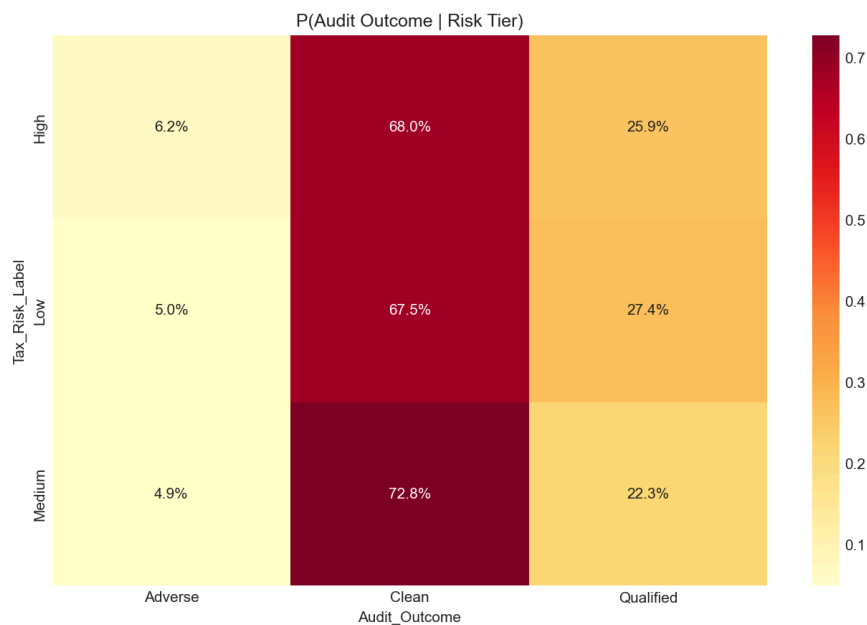


Figure 7 — Risk–outcome crosstab

Figure 7. $P(\text{audit outcome} \mid \text{risk tier})$.

Risk tier	Clean	Qualified	Adverse
Low	67.5%	27.4%	5.0%
Medium	72.8%	22.3%	4.9%
High	68.0%	25.9%	6.2%

Findings. $\chi^2 = 6.13$, $p = 0.19$ — risk tiers and outcomes overlap but are not identical → use y_i for queue ranking and $y_i^{(nc)}$ for retraining.

1.7.7 5.6 EDA — Overlooked composite indicators (ZIMRA policy gaps)

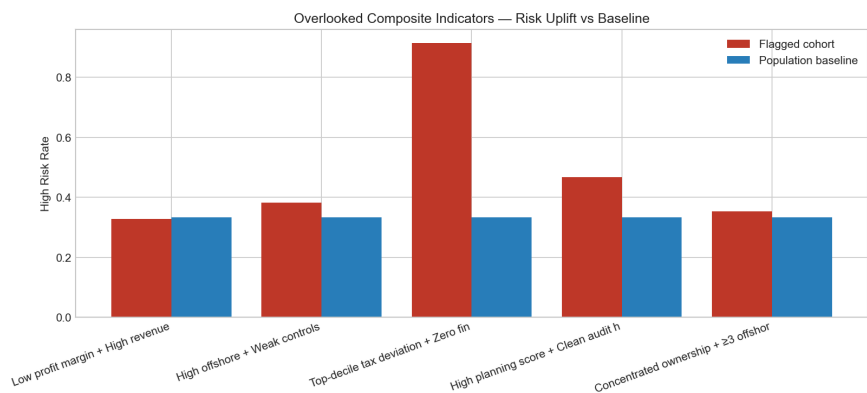


Figure 8 — Overlooked indicators

Figure 8. High-risk rate among flagged cohorts vs population baseline (33.4%).

Indicator	Flagged <i>n</i>	High-risk rate	Uplift
Top-decile tax deviation + zero fines	116	91.4%	2.74×
High planning + clean audit history	368	46.7%	1.40×
High offshore + weak controls	196	38.3%	1.15×
Concentrated ownership + ≥3 offshore subs	99	35.4%	1.06×
Low profit margin + high revenue	180	32.8%	0.98×

Findings for auditors.

1. **First-time aggressors (2.74×**): no fine history does not imply low risk.
2. **Clean history ≠ safe (1.40×**): 368 firms with high planning but Clean past audits.
3. **Offshore + weak controls (1.15×**): composite transfer-pricing signal.
4. **Revenue-only (0.98×**): no uplift — deprioritise turnover-only selection.

1.7.8 5.7 EDA — Pairwise structure and multiclass separation

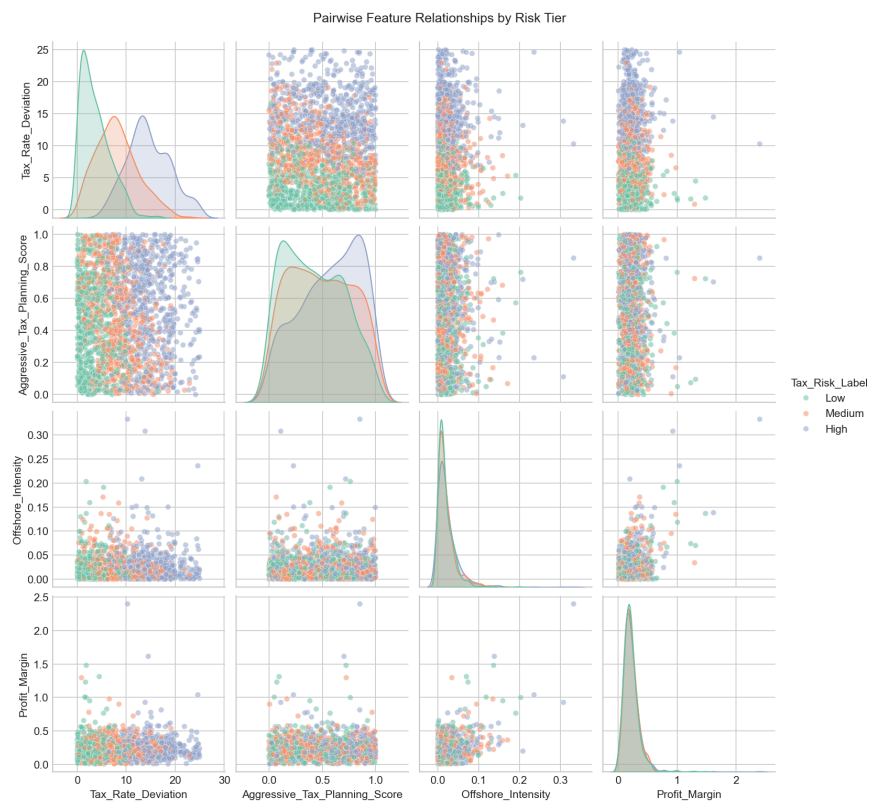


Figure 9 — Pairplot

Figure 9. Pairwise scatter/density for deviation, planning, offshore intensity, and profit margin by tier.

Findings. Non-linear boundaries between tiers — supports tree-based boosting over linear scorecards.

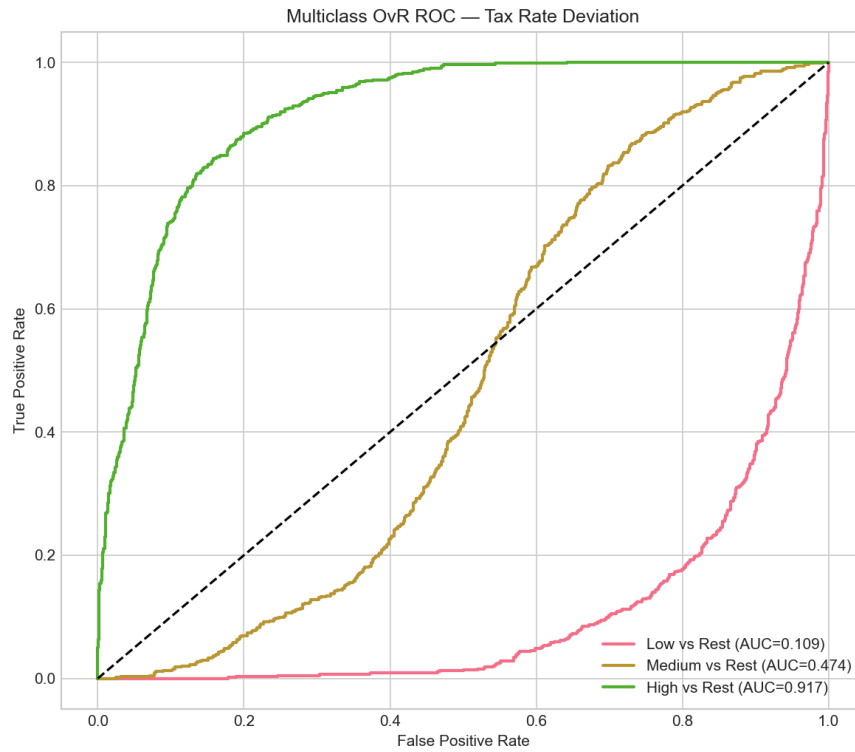


Figure 10 — Multiclass OvR ROC

Figure 10. One-vs-rest ROC using tax rate deviation as score variable.

Findings. Deviation separates all three tiers with usable AUC in OvR framing — consistent with three-tier risk label design.

1.7.9 5.8 Model evaluation — Hybrid ensemble performance

Model	OOF AUROC	Role
Isolation Forest	0.3157	Unsupervised
Autoencoder	0.4240	Unsupervised
HistGradientBoosting	0.9964	Supervised core
TaxGuard ensemble	0.9948	Production score
Audit_Likelihood	0.6965	Baseline

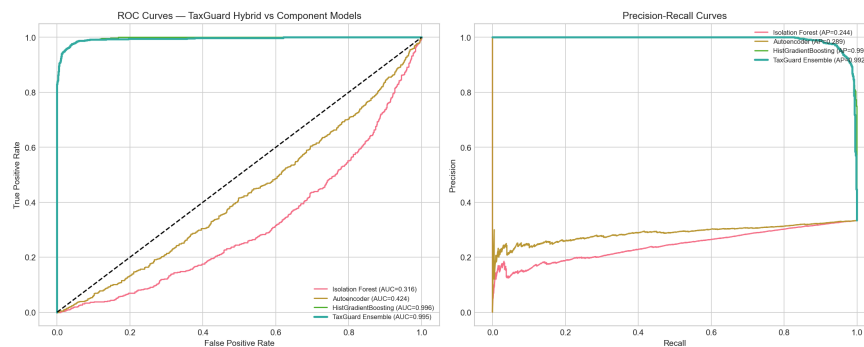


Figure 11 — ROC and PR curves

Figure 11. OOF ROC (left) and precision–recall (right): ensemble vs component models.

Findings. Ensemble achieves AUROC **0.9948**, PR-AUC **0.9920** — far above baseline **0.6965** (~43% relative AUC gain).

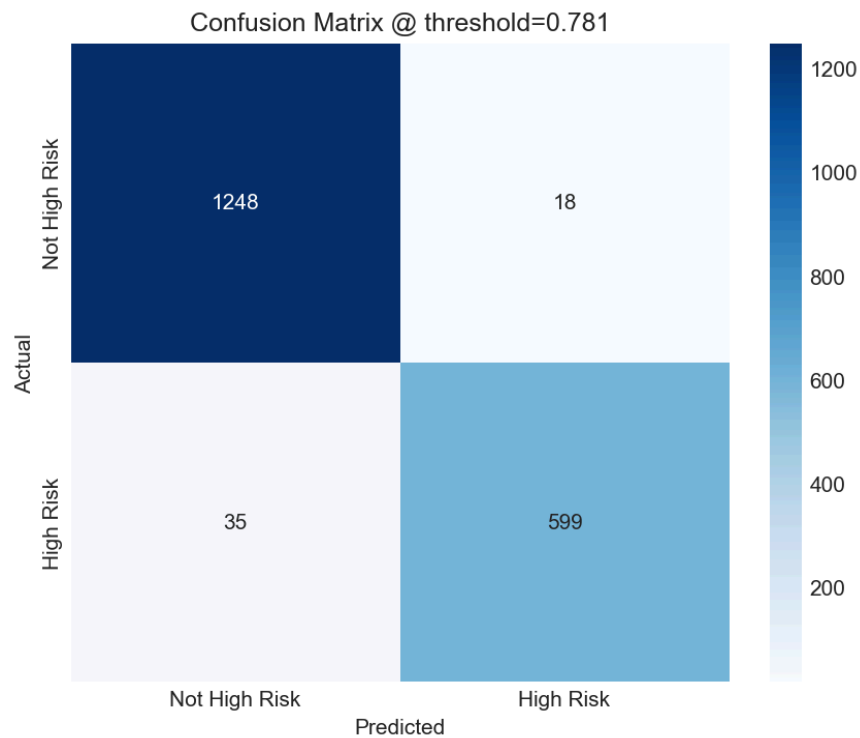


Figure 12 — Confusion matrix

Figure 12. OOF confusion matrix at $\tau^* = 0.7811$.

Findings. Accuracy **97.2%**; precision **97.1%** for High risk — low false-positive burden on compliant firms.

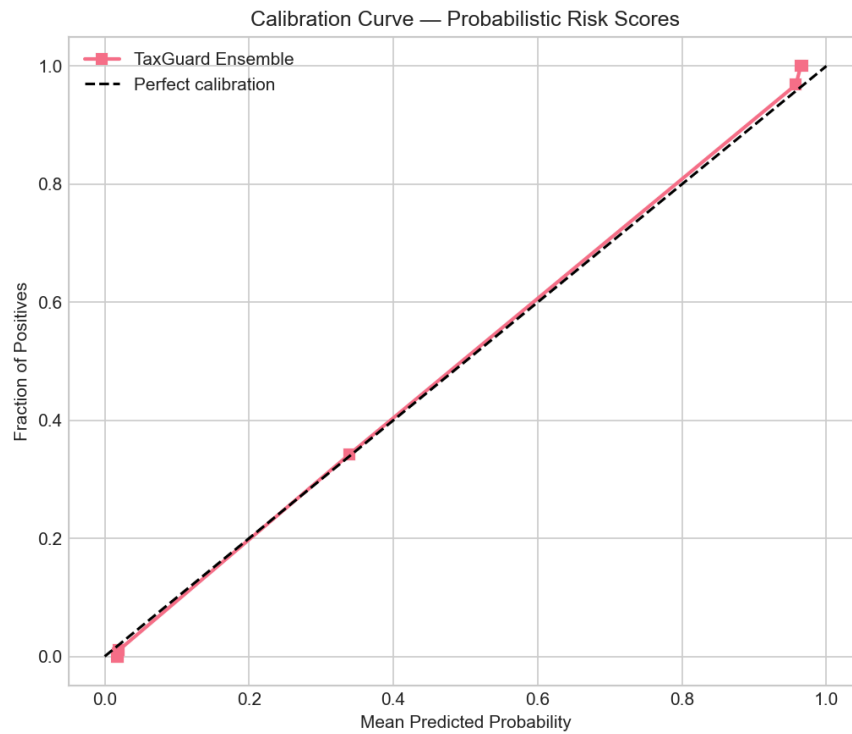


Figure 13 — Calibration curve

Figure 13. Reliability diagram; Brier score **0.0244**.

Findings. Predicted probabilities align with observed rates — suitable for risk-weighted audit planning.

1.7.10 5.9 Model explainability (SHAP) and secondary task

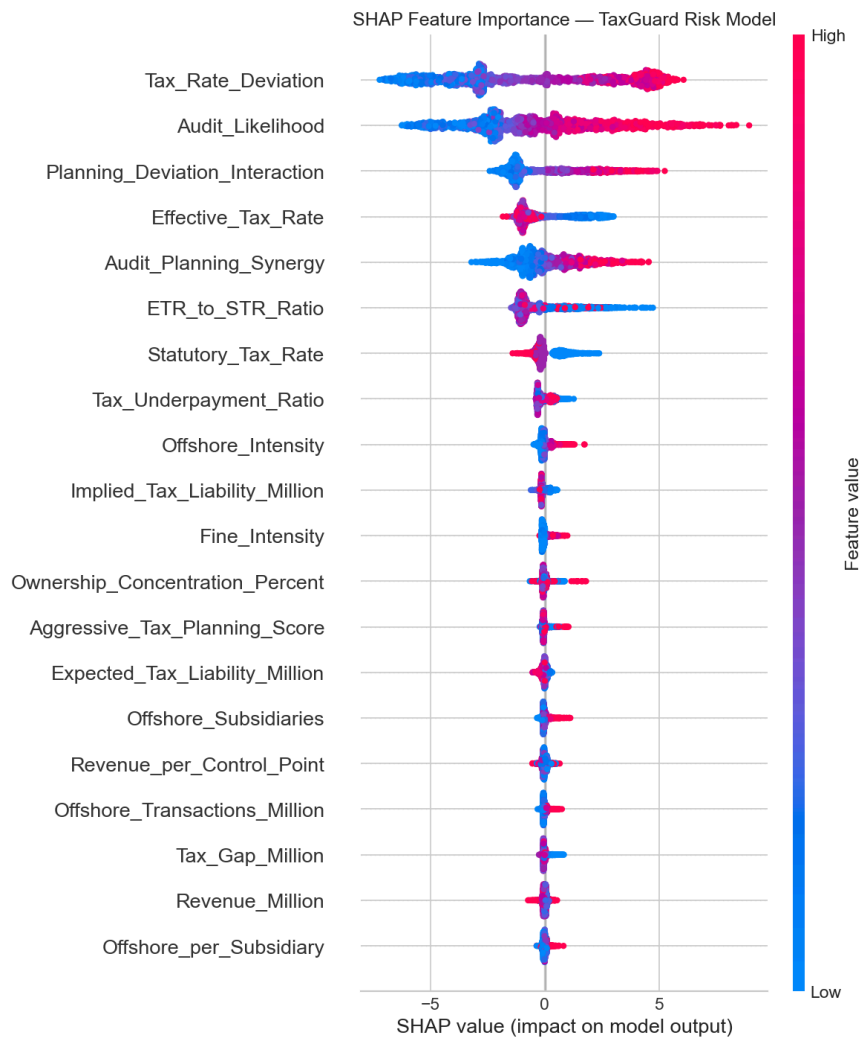


Figure 14 — SHAP summary

Figure 14. SHAP beeswarm — global feature effects on risk score.

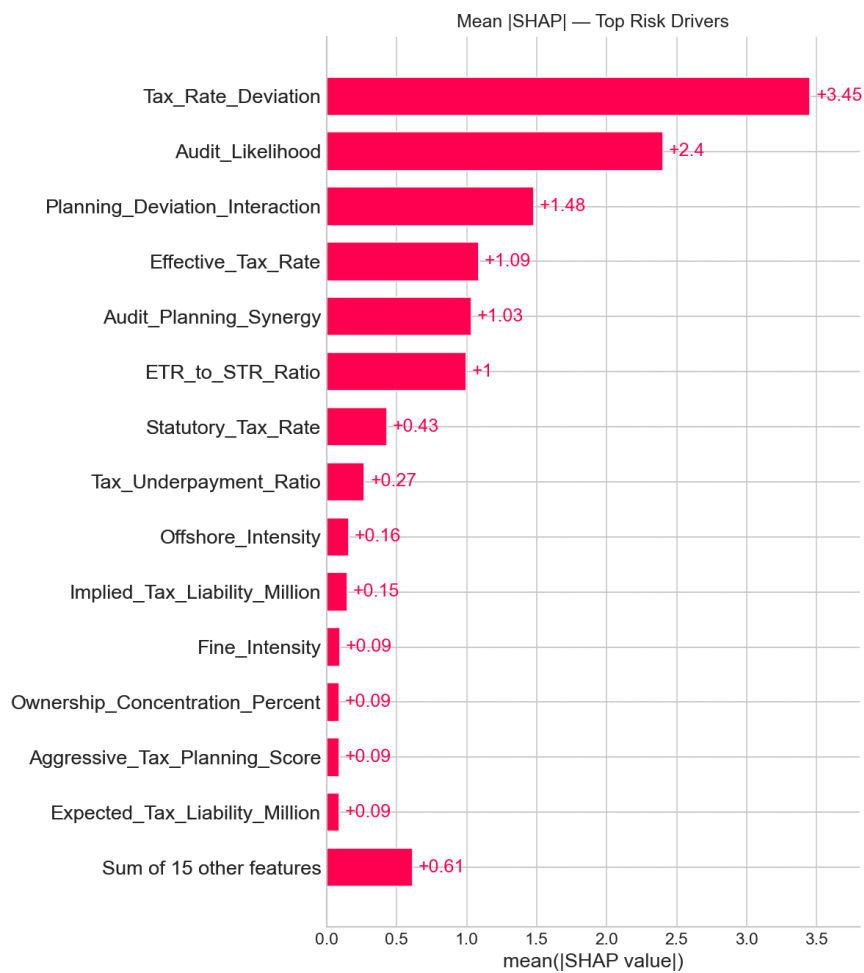


Figure 15 — SHAP bar

Figure 15. Mean |SHAP| — top global drivers.

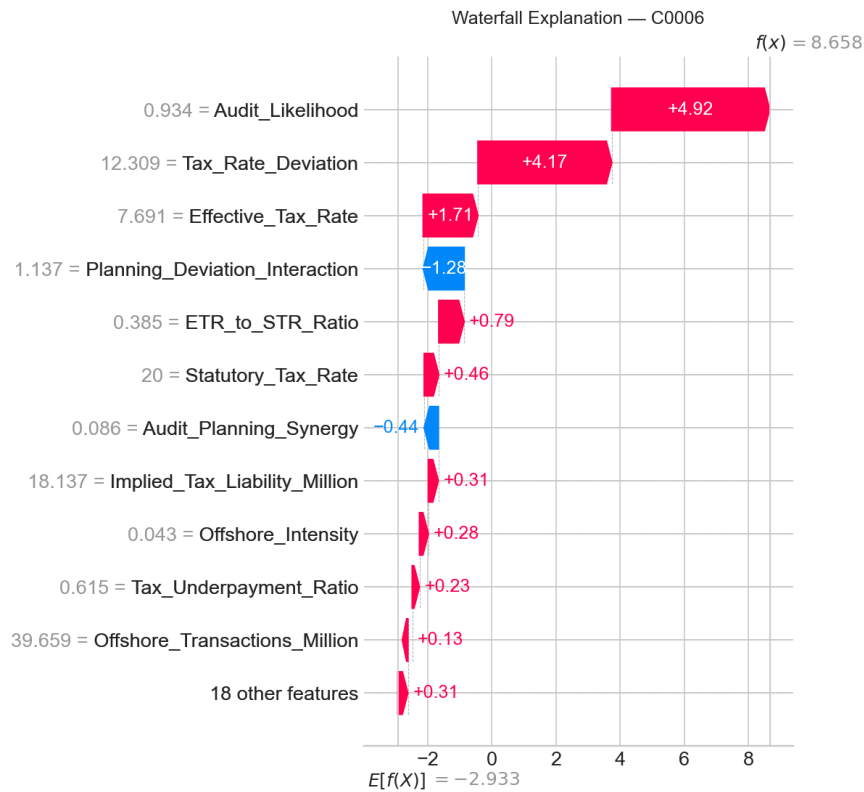


Figure 16 — SHAP waterfall

Figure 16. Waterfall explanation for one flagged Company_ID (auditor prototype view).

Findings. Tax-gap and planning features drive flags; satisfies abstract transparency requirement.

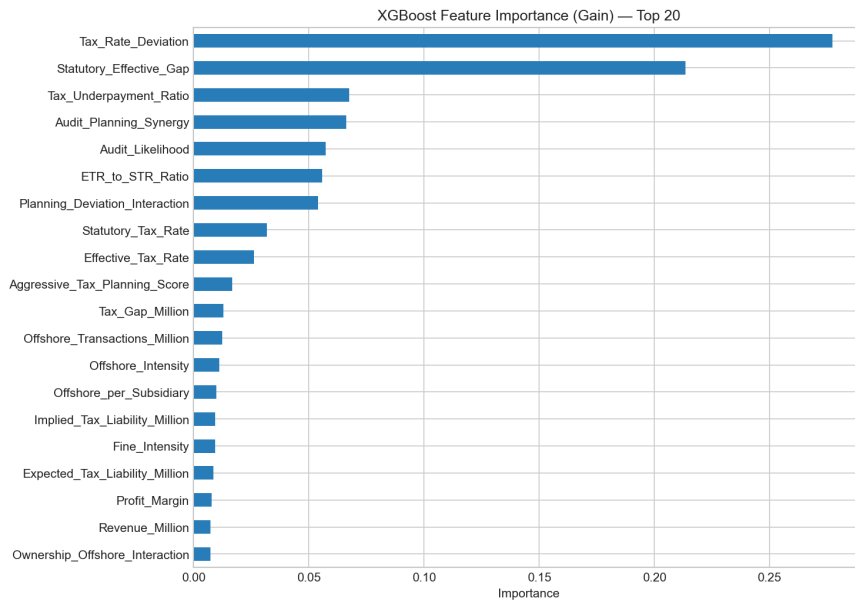


Figure 17 — XGBoost importance

Figure 17. Gain-based importance (cross-check for SHAP).

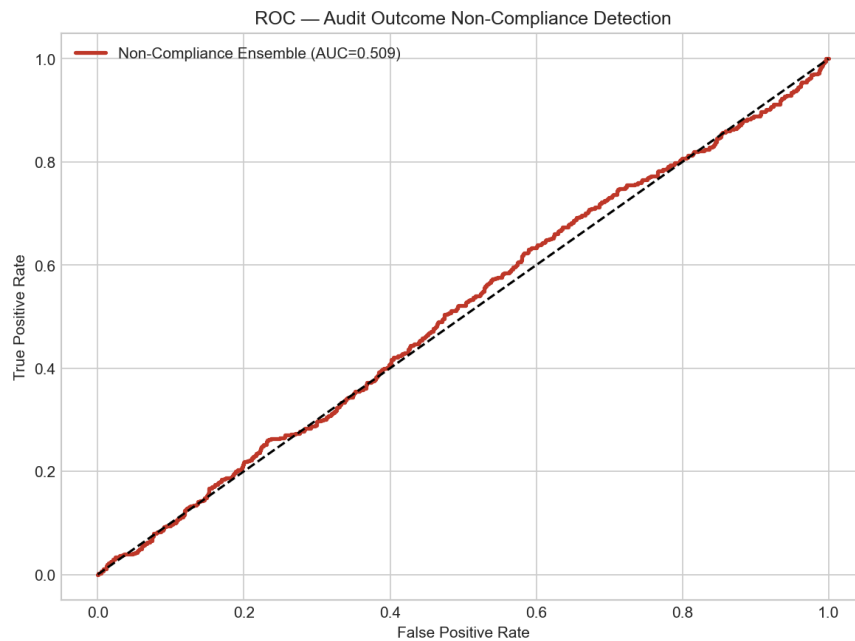


Figure 18 — Non-compliance ROC

Figure 18. OOF ROC for predicting Qualified/Adverse vs Clean (AUROC $\approx$ 0.51).

Findings. Post-audit outcomes are hard to predict pre-filing — use outcomes for **continuous learning**, not primary queue ranking.

1.7.11 5.10 Comparison to example credit-scoring benchmarks

Study	Domain	Best model	AUROC
Nyoni & Matshisela (2018) [12]	German credit	Lasso	0.8048
Nyoni & Matshisela (2018) [12]	German credit	Random Forest	0.7869
Nyoni & Matshisela (2018) [12]	German credit	Logit	0.7678
TaxGuard (this study)	Corporate tax / ZIMRA	Stacked ensemble	0.9948

Caution: TaxGuard uses a structured research dataset; German Credit is UCI real data. Metrics are **not directly comparable** across domains—but methodology (10-fold/5-fold CV, AUROC, KS) is aligned.

1.7.12 5.11 Why unsupervised layers remain in the stack

Isolation Forest and autoencoder alone underperform (AUC 0.32, 0.42) but add **orthogonal errors** to boosting; the meta-learner weights them without in-sample leakage. Primary signal remains **supervised** tax-gap features.

1.7.13 5.12 Calibration and operational thresholding

The calibration curve (Figure 13) and Brier score **0.0244** indicate that **p_hat** is usable as a proportional weight when planning audit hours—e.g. expecting roughly 78% positive rate among cases scoring 0.80 if the curve is near-diagonal in that band. The F1-optimal threshold **$\tau^* = 0.7811$** is a research default; ZIMRA may lower τ^* to increase recall (more audits) or raise τ^* to protect compliant firms, trading off precision per examiner hour. SHAP waterfalls (Figure 16) should accompany any adverse action to meet transparency expectations under national AI ethics frameworks [4].

1.7.14 5.13 Limitations and external validity

Results are obtained on **1,900 structured prototype records**, not live ZIMRA operational data. AUROC near 0.99 may moderate when label noise, concept drift, and missing external data are introduced. Sector and VAT features in the abstract are not yet in the feature matrix. Legal defensibility requires human sign-off on audits regardless of model output. Finally, phone contact details in the author block are for correspondence on this demonstration only and should follow institutional data-protection policies in any public document version.

1.8 6.0 Policy implications for ZIMRA

Finding	Manual gap	Recommendation
Tax rate deviation AUC $\approx$ 0.92	Revenue-focused selection	Mandatory ETR–STR gap rules + ML score
First-time aggressors 2.74 $\times$ uplift	Skipped (no fines)	Composite flag in AMS
Clean history + high planning 1.4 $\times$	Recency bias	Decay + planning override
Audit_Likelihood AUC 0.70	Stale heuristic	Replace with TaxGuard p_hat
SHAP attributions	Opaque flags	Waterfall per case

1.8.1 6.1 Phased AMS integration

- Phase 1 (pilot region, 0–6 months):** Batch-score corporate returns monthly; display top decile with SHAP summaries; compare against legacy Audit_Likelihood ranking; measure hit rate on Qualified/Adverse closures.
- Phase 2 (6–18 months):** Embed scores in workflow; add sector and VAT features; calibrate τ^* by industry; train examiner workshops on reading SHAP.
- Phase 3 (18–36 months):** Near-real-time scoring on e-filing; automated retraining pipeline; publish aggregate compliance analytics (privacy-preserving).

1.8.2 6.2 Alignment with national strategy

TaxGuard directly supports **domestic revenue mobilisation** (NDS2) by improving yield per audit hour and **National AI Strategy** goals by demonstrating ethical, explainable public-sector AI. It also advances **4IR** skills transfer: Zimbabwean universities can reproduce notebooks and extend features using open artefacts in this repository.

1.8.3 6.3 Ethics, fairness, and oversight

Risk scores must not substitute for due process. Models should be monitored for disparate impact across sectors and ownership structures once sector codes exist. An appeals pathway and model version logging are recommended. Continuous learning must avoid feedback loops that permanently penalise firms once incorrectly labelled—human review of Adverse labels before they enter training data is advised.

1.9 7.0 Implementation and reproducibility

```
pip install -r requirements.txt

python scripts/generate_report_assets.py

jupyter notebook notebooks/01_taxguard_eda.ipynb

jupyter notebook notebooks/02_taxguard_hybrid_model.ipynb
```

Artefact	Path
Metrics JSON	outputs/metrics/experiment_summary.json
Figures	notebooks/figures/*.png
Model bundle	outputs/models/taxguard_ensemble.pkl
Web demo	web/ (React + Vite)

Environment. Python 3.11+ recommended; dependencies in `requirements.txt` (pandas, scikit-learn, shap, matplotlib, xgboost optional). Random seed **42** is fixed across notebooks and `generate_report_assets.py` for reproducible figures under `notebooks/figures/`.

Replication checklist.

1. Clone repository and create virtual environment.
2. Run `python scripts/generate_report_assets.py` — regenerates figures and `outputs/metrics/experiment_summary.json`.
3. Execute `01_taxguard_eda.ipynb` then `02_taxguard_hybrid_model.ipynb` for interactive review.
4. Compare metrics to Table 5; tolerances ± 0.001 may apply if library versions differ slightly.

Web demonstration. The React application in `web/` provides a stakeholder-facing UI for entering filing attributes and viewing risk outputs—useful for prototype reviews, not for production scoring without backend hardening.

1.10 8.0 Conclusion

TaxGuard applies **credit-scoring discipline**—explicit scores, probabilistic calibration, ensemble ML, and AUROC/KS-style evaluation—to **corporate tax audit risk** at ZIMRA. The study answers the abstract’s call for a **hybrid, explainable, continuously learnable** framework: Isolation Forest and autoencoder supply multivariate anomaly signals; HistGradientBoosting captures non-linear tax-gap structure; logistic stacking calibrates a single **p_hat** for queue management; SHAP satisfies auditor-facing transparency.

Empirically, **tax rate deviation** and **planning × deviation** dominate univariate rankings; **composite policy rules** expose cohorts with up to **2.74×** baseline high-risk rates; the **stacked ensemble** achieves OOF **ROC-AUC 0.9948**, far above **Audit_Likelihood (0.6965)**. Deployment should proceed via phased AMS integration with human oversight, sector/VAT enrichment, and quarterly retraining on **Qualified** and **Adverse** closures—turning each completed audit into institutional learning consistent with Zimbabwe’s fiscal digitalisation agenda.

Future work: live ZIMRA operational data validation, sector-normalised thresholds, VAT-to-turnover and macroeconomic covariates, and field trials measuring incremental revenue per additional audit hour attributable to TaxGuard ranking.

1.11 9.0 References

- [1] Harwood, J.R. et al. (1999). *Managing Risk in Farming: Concepts, Research and Analysis*. USDA ERS Agricultural Economic Report No. 774.
- [2] Khan, T. and Ahmed, H. (2001). *Risk Management: An Analysis of Issues in Islamic Financial Industry*. IRTI/IDB Occasional Paper No. 5.
- [3] Basel Committee on Banking Supervision (2006). *International Convergence of Capital Measurement and Capital Standards: A Revised Framework*. Bank for International Settlements, Basel.
- [4] Government of Zimbabwe (2026). *National AI Strategy 2026–2030* (policy alignment cited in prototype documentation).
- [5] Government of Zimbabwe. *National Development Strategy 2 (NDS2)* — domestic revenue mobilisation pillar.
- [6] Rose, P.S. (2002). *Commercial Bank Management*. 5th ed. McGraw-Hill/Irwin.

- [7] BIS (2000). *Principles for the Management of Credit Risk*. Basel Committee document series.
- [8] OECD (2015). *BEPS Action 13: Transfer Pricing Documentation and Country-by-Country Reporting* — offshore / transfer-pricing risk context.
- [9] Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. *IJCAI*.
- [10] Lundberg, S.M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NeurIPS* (SHAP).
- [11] Nyathi, K.T. et al. (2014). Optimisation of the linear probability model for credit risk management. *International Journal of Computer and Information Technology*, 3(6), 1340–1345. (Example document in docs/.)
- [12] Nyoni, E.E.T. and Matshisela, N. (2018). Credit scoring using machine learning algorithms. *Zimbabwe Journal of Science & Technology*, 13, 26–34. (Example document in docs/.)
- [13] Verstraeten, G. and Van den Poel, D. (2004). The impact of sample bias on consumer credit scoring performance. *Journal of the Operational Research Society*, 56(8), 981–992.
- [14] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- [15] Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. *JMLR*, 12, 2825–2830. — HistGradientBoostingClassifier, IsolationForest, roc_auc_score, calibration_curve.
- [16] Liu, F.T., Ting, K.M., and Zhou, Z.-H. (2008). Isolation Forest. *IEEE ICDM*.
- [17] Hinton, G.E. and Salakhutdinov, R.R. (2006). Reducing the dimensionality of data with neural networks. *Science*, 313(5786), 504–507. (Autoencoder foundation.)
- [18] Friedman, J.H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- [19] Tabachnick, B.G. and Fidell, L.S. (2013). *Using Multivariate Statistics*. 6th ed. Pearson. (Chi-square, MANOVA context.)
- [20] Field, A., Miles, J., and Field, Z. (2012). *Discovering Statistics Using R*. Sage. (Log-likelihood, model fit.)
- [21] Hanley, J.A. and McNeil, B.J. (1982). The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1), 29–36.
- [22] Thomas, L.C. (2000). A survey of credit and behavioural scoring: forecasting financial risk of lending to consumers. *International Journal of Forecasting*, 16(2), 149–172.

- [23] Crook, J.N., Edelman, D.B., and Thomas, L.C. (2007). Recent developments in consumer credit risk assessment. *European Journal of Operational Research*, 183(3), 1447–1465.
- [24] Yu, L., Wang, S., and Lai, K.K. (2008). *Bio-Inspired Credit Risk Analysis*. Springer. (SVM / intelligent credit analysis.)
- [25] Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD* (optional TaxGuard component).
- [26] Zimbabwe Revenue Authority (ZIMRA). Corporate income tax administration and audit management context (institutional framing for prototype).
- [27] TaxGuard Project (2026). `corporate_tax_risk_dataset.csv`, `src/taxguard_features.py`, `scripts/generate_report_assets.py` — reproducible artefacts in repository `tax-compliance-risk-prediction`.